C 62652

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Name.....

Reg. No.....

SECOND SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION MAY 2019

B.Sc. Statistics

STS 2C 02—PROBABILITY DISTRIBUTIONS

Time : Three Hours

Maximum : 80 Marks

Section A

Answer **all** *questions in* **one word.**
Each question carries 1 *mark.*

Name the following :

1. The second central moment of a random variable X.
2. $E\left[e^{itX}\right]$ of a random variable X.
3. Probability distribution of the random variable X denoting the number of failures before the first success in an experiment with only two possible results success and failure.

Fill up the blanks :

4. Pearson's co-efficient of correlation between two random variables satisfying the linear relation $2x + 3y - 5 = 0$ perfectly is ———.
5. Probability density function of X following U [2, 5] = ———.
6. X and Y are two random variables with bivariate m.g.f. function, $M_{X,Y}(t_1, t_2)$ is expressed in terms of expectation as ———.
7. The range of variation of beta distribution of second kind is ———.

Write True *or* False :

8. m.g.f. exists for all the random variables.
9. Normal distribution is symmetric about its median.
10. Central limit theorem discusses the convergence of sum of random variables to normal distribution.

(10 × 1 = 10 marks)

Section B

Answer **all** *questions in* **one sentence**.
Each question carries 2 *marks.*

11. Find the mean of a random variable X with first moment about 4 is given as 7.

12. $f(x, y)$ is the joint p.d.f. of two continuous random variables X and Y. Write any *two* of its properties.

13. Write the mean and variance of a binomial distribution with parameters $(15, 0.3)$.

14. Define covariance of X and Y.

15. If $p(x, y) = \frac{(2x + y)}{10}$, where $x = 0.1$ and $y = 1.2$ is a joint p.m.f. of X and Y, write the p.m.f. of X.

16. State Bernoulli's law of large numbers.

17. Define lognormal distribution.

(7 × 2 = 14 marks)

Section C

Answer any **three** *questions.*
Each question carries 4 *marks.*

18. First three raw moments of X are – 1 , 55 and – 62.5. Obtain co-efficient of skewness based on moments.

19. State and prove Cauchy-Schwartz inequality.

20. For a random variable X, $P(X = 3) = 2P(X = 4) = P(X = 5)$. Find V (X).

21. Obtain the m.g.f. of X following Poisson distribution with parameter λ.

22. The repair time of a machine follows exponential distribution with an average of 2 hours. What is the probability that the repair time will be more than 2 hours ?

(3 × 4 = 12 marks)

Section D

Answer any **four** *questions.*
Each question carries 6 *marks.*

23. $f(x, y) = e^{-x-y}, x > 0, y > 0$ be the joint p.d.f. of (X, Y). Find the bivariate m.g.f. of (X, Y) and hence show that X and Y are independent.

24. Show that $E(XY) = E(X)\,E(Y)$ need not imply X and Y are independent.

25. For two random variables X and Y, the joint p.d.f.

$$f(x, y) = 2 - x - y; 0 \le x \le 1, 0 \le y \le 1$$
$$= 0, \text{ otherwise}$$

. Find E (X) and E (Y).

26. If $X \sim N(\mu_1, \sigma_1)$, $Y \sim N(\mu_2, \sigma_2)$, obtain the distribution of $aX + bY$ when X and Y are independent.

27. If $X \sim N(12, 4)$. Obtain (i) $P(X \le 20)$; (ii) $P(0 \le X \le 24)$; and (iii) $P(|X - 12| \ge 8)$.

28. State and prove Tchebychev's inequality.

$(4 \times 6 = 24$ marks)

Section E

Answer any **two** *questions.*
Each question carries 10 *marks.*

29. Let X and Y are two random variables with joint p.d.f. $f(x, y) = x + y; 0 < x < 1; 0 < y < 1$. Compute the coefficient of correlation between X and Y.

30. Obtain the m.g.f. of X following binomial distribution with parameters n and p. Hence state and prove the additive property of binomial distribution. If X and Y are independent binomial random variables with parameters (6, 0.5) and (4, 0.5) respectively, calculate $P(X + Y \ge 3)$.

31. If $X \sim (\mu, \sigma)$, prove that (a) $\frac{X-\mu}{\sigma} \sim N(0,1)$; (b) The quartile deviation of X is $0.6745\,\sigma$.

32. State and prove Weak law of large numbers. Examine whether WLLN hold good for the sequence of random variables $\{X_i\}, i = 1, 2, \ldots$ where $P\left(X_i = \pm\sqrt{2i-1}\right) = 0.5$.

(2 × 10 = 20 marks)